

# Improved attack detection in IoT and IIoT networks using attention mechanisms in convolutional neural networks

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## ABSTRACT

The widespread adoption of devices on the Internet of Things (IoT) and Industrial Internet of Things (IIoT) in consumer and industrial applications has introduced numerous cybersecurity challenges, with vulnerabilities to sophisticated attacks posing critical operational and financial risks. This study utilizes the latest Edge\_IIoTset dataset, which contains data from IoT and IIoT environments, categorized into normal instances and 14 different types of cyberattack. To enhance attack detection accuracy, we propose a novel deep learning approach that integrates Convolutional Neural Network (CNN) architectures, ResNet50 and MobileNet, with a two-stage attention mechanism. The raw dataset is preprocessed by transforming numerical data into image representations through quantile transformation, enabling the extraction of spatial features. In the first stage, channel attention is used to prioritize the most informative feature channels. In the second stage, spatial attention is applied to the resultant feature channels to focus on important regions within these representations. By incorporating these attention mechanisms into ResNet50 and MobileNet, the models dynamically adapt to highlight critical attack patterns, improving sensitivity to diverse attack types. In addition, quantization techniques are employed to optimize model size, striking a balance between computational efficiency and detection performance. Experimental results reveal that our proposed approach significantly improves detection accuracy, demonstrating the effectiveness of attention-augmented deep learning in securing IoT and IIoT ecosystems against cyberattacks. Key metrics such as inference time, compression ratio, resource utilization, computational efficiency, and computational complexity demonstrate the scalability and suitability of the model for resource-constrained environments.

## 1. Introduction

The rise of the Internet of Things (IoT) and Industrial Internet of Things (IIoT) has revolutionized sectors ranging from manufacturing and logistics to healthcare and energy management. By integrating a vast network of connected devices, these systems facilitate real-time data acquisition and enable autonomous decision-making, leading to smarter, more efficient operations. However, this connectivity introduces significant cybersecurity challenges. With an increasing number of devices communicating across open networks, IoT and IIoT environments are highly susceptible to cyberattacks, which can disrupt industrial processes, compromise sensitive information, and even lead to substantial financial and operational losses. As these devices often lack robust, built-in security protocols, malicious actors can exploit their

vulnerabilities, executing a multitude of cyberattacks including Denial of Service (DoS), data injection, and command-and-control attacks (Kolias et al., 2017).

The development of effective Intrusion Detection Systems (IDS) for IoT and IIoT networks is crucial for safeguarding these environments. Traditional IDS approaches, while useful, are often inadequate in handling the vast and complex datasets generated by IoT devices. Conventional machine learning algorithms, such as Decision Trees, Support Vector Machines, and k-Nearest Neighbors, have been applied to IoT security (Hasan et al., 2019), but they generally lack scalability and struggle to detect complex, multi-dimensional attack patterns inherent in IoT/IIoT networks. Recent advancements in deep learning (DL) have presented more effective alternatives, with architectures like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs)

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showing superior performance in processing large, high-dimensional data sets due to their ability to learn complex feature hierarchies (Shone et al., 2018; Yin et al., 2017).

To support these claims, several studies have demonstrated the empirical superiority of deep learning approaches in IoT security tasks. For example, Yin et al. proposed RNN-IDS with a detection accuracy of 97.09% using a recurrent neural network on the NSL-KDD dataset, significantly outperforming traditional models such as Decision Trees and Support Vector Machines (Yin et al., 2017). In a similar way, Shone et al. employed a stacked non-symmetric deep auto-encoder (S-NDAE) model achieving high detection rate of 97.85% on KDD Cup '99 dataset, demonstrating its effectiveness in capturing complex attack behaviors compared to classical techniques (Shone et al., 2018). These studies reinforce the suitability of deep learning for handling nonlinear, high-volume data characteristic of IoT and IIoT systems.

However, a critical gap remains in the current state-of-the-art IDS models. Most existing deep learning-based IDS models either overlook or insufficiently exploit the spatial structure and internal correlations within transformed IoT data. They typically treat all input features equally, which can dilute the model's focus on attack-relevant regions, particularly in noisy or imbalanced data scenarios. Moreover, few approaches leverage attention mechanisms to dynamically prioritize informative features or spatial cues. These shortcomings limit the detection accuracy and generalization of IDS models in real-world IoT/IIoT environments. Therefore, there is a pressing need for advanced models that can effectively concentrate on the most relevant parts of the input data and extract discriminative patterns for accurate threat identification.

Attention mechanisms have emerged as a propitious solution to this problem, enabling models to selectively focus on significant aspects of the input data. By integrating attention mechanisms, particularly channel and spatial attention, it is possible to enhance a model's ability to prioritize the most relevant features or spatial regions within the data (Guo et al., 2022). In recent research, attention mechanisms have shown impressive results in diverse domains, which includes medical imaging and natural language processing, where they help models to achieve higher accuracy by focusing on critical patterns within high-dimensional inputs (Vaswani et al., 2017; Xie et al., 2023). This selective focus is especially valuable in IoT/IIoT environments where attack patterns are often hidden within a vast amount of normal operational data.

This study explores the use of the Edge\_IIoTset dataset, a comprehensive dataset containing normal data and multiple types of attack data in both IoT and IIoT contexts. To improve the feature extraction, we transform the dataset into image representations using quantile transformation, allowing deep learning models to interpret the data as spatial structures and thereby enabling effective feature extraction. We use ResNet50 and MobileNet as base architectures for attack detection due to their well-established capabilities in image recognition tasks. ResNet50's residual learning framework mitigates the vanishing gradient problem, allowing deeper networks to learn complex features, while MobileNet's lightweight architecture is particularly suitable for deployment in resource-constrained IoT devices.

To further enhance these models, we apply channel and spatial attention mechanisms, which help the model to learn which features and regions are most indicative of an attack. Channel attention emphasizes the most relevant feature channels, effectively weighting features such as device behavior or network traffic attributes, while spatial attention focuses on the most important regions within an image. By guiding the model's focus, these mechanisms improve the ability to detect subtle attack patterns, even in complex, noisy environments typical of IoT and IIoT networks. Through extensive experimentation, we demonstrate that attention-enhanced versions of ResNet50 and MobileNet significantly improve detection accuracy over standard deep learning approaches.

The key contributions of this study are outlined as follows:

- **Image-Based transformation for attack detection:** We put forward a novel approach by transforming IoT/IIoT data into image

representations using quantile transformation, facilitating spatial feature extraction by deep learning models.

- **Integration of attention mechanisms:** We apply channel and spatial attention mechanisms to ResNet50 and MobileNet models, enabling them to focus on critical features and regions that are indicative of attacks, thereby improving detection accuracy.
- **Comprehensive evaluation on Edge\_IIoTset:** We rigorously evaluate our attention-augmented models on the Edge\_IIoTset dataset, demonstrating superior performance in detecting a variety of attack types compared to baseline models without attention.
- **Improved accuracy in IoT/IIoT attack detection:** Our findings provide valuable insights for enhancing the security of IoT and IIoT networks, showing that attention-enhanced deep learning models can significantly improve attack detection performance.

The rest of the paper is structured as follows: Section 2 provides a concise review of the latest advancements in attack detection for IoT and applying attention mechanisms within CNNs. Section 3 introduces the proposed framework, dataset used, preprocessing stages, data transformations, channel and spatial attention mechanisms, deep transfer learning (DTL), selected CNN architectures, and model quantization. Section 4 outlines the experimental setup and performance evaluation metrics. Section 5 presents the results and discussion, assessing the model's performance and comparing it with existing frameworks. Section 6 discusses future research directions. Section 7 draws an inference from the study, summarizing the key findings and exploring the wider implications of the research.

## 2. Related work

Cyberattack detection in IoT and IIoT environments has been a significant field of study due to the critical security challenges raised by the distributed and resource-constrained nature of these systems. Traditional machine learning techniques such as Random Forest (Zhang et al., 2008), Support Vector Machines (Ahmad et al., 2018) and Decision Trees (Sahu & Mehtre, 2015) have been used extensively for attack detection in IoT systems (Al-Garadi et al., 2020). However, these methods often struggle to handle the complex, high-dimensional data typically generated by IoT networks (Diro & Chilamkurti, 2018). Deep learning (DL) methods have shown promise in addressing these challenges by capturing intricate patterns in high-dimensional data. For example, Shone et al. utilized stacked autoencoders for network attack detection (Shone et al., 2018), and Yin et al. applied Recurrent Neural Networks (RNN) for attack detection in network traffic (Yin et al., 2017). While these DL models have improved detection capabilities, they lack a mechanism for selectively prioritizing features, which is essential for capturing subtle attack patterns in noisy data.

Awajan developed a fully connected deep neural network (DNN) with four layers for attack detection in IoT environments. This architecture was optimized for resource-constrained IoT devices, using techniques such as hyperparameter tuning and regularization to prevent overfitting. The model was trained on the NSL-KDD dataset, a popular benchmark for network attack detection, and achieved a detection accuracy of 93.21%, illustrating its suitability for IoT-specific challenges (Awajan, 2023).

Yang et al. proposed an attack-traffic-identification framework utilizing contrastive learning to improve accuracy by learning semantic feature representations from unlabeled data. The approach incorporates a Transformer-based feature extraction model with a self-attention mechanism to analyze malicious traffic byte features, alongside a bidirectional GELU Long Short-Term Memory (GLSTM) for capturing temporal patterns. Experimental results demonstrated the method's accuracy of 99.48% when assessed on the BoT-IoT dataset, offering an effective solution for malicious traffic detection (Yang et al., 2023).

Recent research has increasingly turned to Transformer-based architectures due to their capability to model long-range dependencies

and capture contextual patterns effectively in network traffic data. For instance, TranAD, a Transformer-based anomaly detection model, employs dual-stage attention-based encoding to capture both short and long-range temporal dependencies and has demonstrated state-of-the-art performance on industrial IoT datasets such as SWaT and WADI (Tuli et al., 2022). Similarly, LogBERT, a self-supervised model based on Bidirectional Encoder Representations from Transformers (BERT), learns normal log patterns and effectively detects anomalies in system logs, outperforming traditional RNN-based approaches on benchmark datasets like HDF5, BGL and Thunderbird (Guo et al., 2021).

Graph-based learning has also shown promise in the intrusion detection landscape. Graph Convolution Networks (GCNs) and their variants leverage the structural information inherent in network traffic and device relationships. Sun et al. developed GNN-IDS, a graph neural network (GNN) based IDS where each network flow is represented as a node in a graph, enabling the model to detect complex, distributed attack patterns. The model achieved performance with a detection accuracy of 94.75 % on CIC-IDS2017 dataset, showcasing the adaptability of GNNs to IoT attack surfaces (Sun et al., 2024). Moreover, Wai et al. proposed E-GraphSAGE, a GNN model designed to capture both edge features and topological information within a graph structure for enhanced network intrusion detection in IoT networks. By leveraging neighborhood aggregation alongside rich edge representations, the model effectively improved real-time traffic classification performance and demonstrated high accuracies of 81 % and 87 % on the NF-BoT-IoT and TON\_IoT datasets, respectively (Lo et al., 2022). These modern approaches highlight the evolving landscape of intrusion detection beyond traditional CNNs and RNNs.

Alkadi et al. put forward a Collaborative Intrusion Detection System (CIDS) that leverages blockchain and IDS to enhance attack detection in IoT networks. The Deep Blockchain Framework (DBF) integrates distributed attack detection with privacy-preserving blockchain technology for securing IoT networks and supporting the live migration of virtual machines (VMs) in cloud environments. By sharing attack intelligence among nodes, their system improved the ability to detect both localized and distributed attacks. This collaborative approach combined machine learning algorithms with real-time data sharing to achieve greater detection accuracy. The framework was evaluated using the UNSW-NB15 dataset and achieved an accuracy of approximately 97.26 %, demonstrating its effectiveness over standalone IDS systems (Alkadi et al., 2020).

Recent progress in deep transfer learning (DTL) has significantly contributed to the development of effective cybersecurity solutions for IoT environments. For instance, Bierbrauer et al. proposed a transfer learning-based intrusion detection system that analyzes raw network traffic to identify threats. Their approach combined a one-dimensional CNN with a retained random forest classifier and was trained on the UNSW-NB15 and CICIDS2017 datasets, achieving an accuracy of 96.89 % (Bierbrauer et al., 2023).

Rashid et al. introduced a federated learning (FL) approach for intrusion detection that preserves data privacy by training models locally on IoT devices and sharing only model updates with a central server. This decentralized method enhances security while maintaining performance. Tests on the Edge-IIoTset dataset showed an accuracy of 92.49 %, comparable to the 93.92 % achieved by centralized models (Rashid et al., 2023).

Hussain et al. introduced an IDS based on the ResNet18 architecture, a deep residual network known for its efficient feature extraction capabilities. By incorporating skip connections, the model addressed the vanishing gradient problem, enabling it to capture complex attack patterns effectively. Data preprocessing techniques, including feature scaling, further enhanced its performance. The framework was evaluated on the CICDDoS2019 dataset, attaining a high accuracy of 87 % for multi-class detection, establishing it as one of the most efficient deep learning-based approaches for attack detection in IoT environments (Hussain et al., 2020).

Fan et al. investigated the integration of federated learning (FL) and transfer learning (TL) to build an IDS specially made for 5G IoT networks. Federated learning allowed the model to train collaboratively across distributed IoT devices while taking care of data privacy, and transfer learning facilitated the adaptation of pre-trained models to specific intrusion detection tasks. The study employed CICIDS2017, a popular dataset, NSL-KDD and three other special IoT datasets combining 5G network traffic and IoT attack scenarios, achieving an accuracy of 91.93 % on average. This approach addressed the dual challenges of data privacy and the need for adaptable IDS models in 5G-enabled IoT systems (Fan et al., 2020).

Ge et al. adopted a TL approach for multi-category intrusion detection in IoT environments. The study employed a pre-trained CNN architecture, fine-tuned for IoT traffic patterns. Feature extraction and layer freezing techniques were used to retain the knowledge from the base model while adapting to the new intrusion detection task. Evaluated on the BoT-IoT dataset, the model attained an accuracy of 99.79 %, demonstrating its capacity to handle diverse attack categories effectively (Ge et al., 2021).

Guan et al. proposed a TL framework to fine-tune pre-trained deep learning models for attack detection in IoT networks. Their methodology emphasized the adaptability of transfer learning in scenarios with limited labeled data, significantly reducing training time while maintaining high performance. The study evaluated its model on the USTC-TFC2016 dataset, achieving an average accuracy of 98.91 %, showcasing its ability to effectively detect a range of cyberattacks across different IoT scenarios (Guan et al., 2021).

In cybersecurity applications, CNNs have proven effective for attack detection due to their hierarchical feature extraction capabilities. Architectures such as ResNet50 and MobileNet have been widely adopted in image-based recognition tasks for their ability to capture spatial features, even in high-dimensional data. ResNet50's residual learning framework mitigates the vanishing gradient problem, making it suitable for deep architectures (He et al., 2016), while MobileNet's lightweight design is particularly advantageous for resource-constrained IoT environments (Howard, 2017). However, traditional CNNs often lack the ability to selectively focus on important features or spatial regions, a limitation that attention mechanisms have been designed to address.

Attention mechanisms, particularly channel and spatial attention, have gained prominence in deep learning for their ability to direct a model's focus toward the most informative parts of input data. In computer vision and natural language processing, attention mechanisms have been widely used to improve model performance by enabling selective emphasis on key features (Guo et al., 2022).

Long et al. combined an Adaboost-based Graph Convolutional Network (GCN) with channel attention for IoT attack detection. In this model, channel attention highlighted key features within network traffic, while Adaboost provided an ensemble learning framework to enhance model robustness. This hybrid approach illustrates the advantages of attention mechanisms when applied with GCNs and ensemble methods to increase resilience against diverse types of cyberattacks in IoT systems (Long et al., 2023).

In addition to attack detection, attention mechanisms have been successfully applied in other fields, providing insights relevant to IoT and IIoT security. For instance, Almusawi used a CNN with Convolutional Block Attention Module (CBAM) for fingerprint forgery detection, demonstrating the potential of attention to focus on subtle yet critical features in biometric data (Almusawi et al., 2024).

Several researchers have employed attention mechanisms in combination with hybrid deep learning models to enhance detection accuracy in IoT environments. For instance, Dai et al. presented a CNN-BiLSTM-Attention scheme that amalgamates the robustness of CNNs for feature extraction, BiLSTM for modeling temporal relationships, and an attention layer for concentrating on the most relevant patterns. Their results underscore the potential of attention-enhanced hybrid models in capturing complex attack patterns and enhancing detection accuracy

(Dai et al., 2024). This model demonstrates how hybrid approaches can be particularly effective for IoT environments where temporal and spatial relationships in data need to be considered.

Attention mechanisms have also been explored in non-cybersecurity domains, suggesting their versatility and effectiveness. For example, Wu et al. applied an attention-based CNN for water image classification under IoT settings. Their results demonstrated that attention mechanisms improved the classification performance by selectively emphasizing significant visual features in water images, underscoring the adaptability of attention in IoT environments (Wu et al., 2020). Similarly, Awan et al. used the VGG19 model with spatial attention for image-based malware classification. By employing spatial attention, their model achieved better accuracy by focusing on critical regions within malware images, which can be highly relevant to the spatial structure-based approach used in this study (Awan et al., 2021). Although these studies focus on different applications, their successful use of attention mechanisms to enhance model accuracy provides insights relevant to attack detection in IoT and IIoT.

Our research builds upon the foundation of the above studies, introducing a novel application of channel and spatial attention mechanisms to ResNet50 and MobileNet for enhanced attack detection in IoT and IIoT environments. While prior works have used attention in specific security contexts, our study is unique in that it applies both channel and spatial attention within a CNN framework to high-dimensional image-based transformations of IoT/IIoT data. By transforming the Edge\_IIoTset dataset into image representations, our method leverages CNNs to capture spatial dependencies and selectively focus on the most informative regions. This approach not only enhances the detection accuracy of the models across various types of attacks but also ensures efficiency suitable for resource-constrained IoT devices. Our findings demonstrate that combining CNN-based architectures with channel and spatial attention mechanisms provides a high-performance, scalable solution for real-time IoT/IIoT security applications, contributing an effective tool to the growing field of attention-based cybersecurity solutions. Furthermore, quantized models are more suitable for resource-constrained IoT environments.

### 3. Proposed framework

This section outlines the proposed framework. It starts with an introduction to the dataset, followed by data preprocessing and transformation into a suitable format. Next, the key components of the scheme are discussed, including deep transfer learning (DTL) to leverage pre-trained models and the integration of an attention mechanism to enhance attack detection performance. Finally, Fig. 1 presents a block schematic offering a visual overview of the suggested framework.

#### 3.1. Dataset

Our suggested framework utilizes the Edge\_IIoTset dataset, one of the most recent, pragmatic, and comprehensive cybersecurity datasets available for IoT and IIoT applications. Developed by Ferrag et al. (2022) to simulate a pragmatic IIoT environment, this publicly accessible dataset includes data from over ten types of IoT devices, such as temperature and humidity sensors, ultrasonic and water level detectors, soil moisture sensors, flame and pH sensors, and heart rate monitors. It also encompasses data constituting fourteen different attack classes pertinent to IoT and IIoT network protocols, making it highly valuable for cybersecurity research.

The Edge\_IIoTset dataset was selected for this study due to its rich representation of real-world IoT and IIoT attack scenarios. It includes 14 different types of cyberattacks, covering categories such as Denial of Service (DoS), Man-in-the-Middle (MITM), ransomware, and injection attacks. This variety enables robust evaluation of model performance across multiple threat vectors.

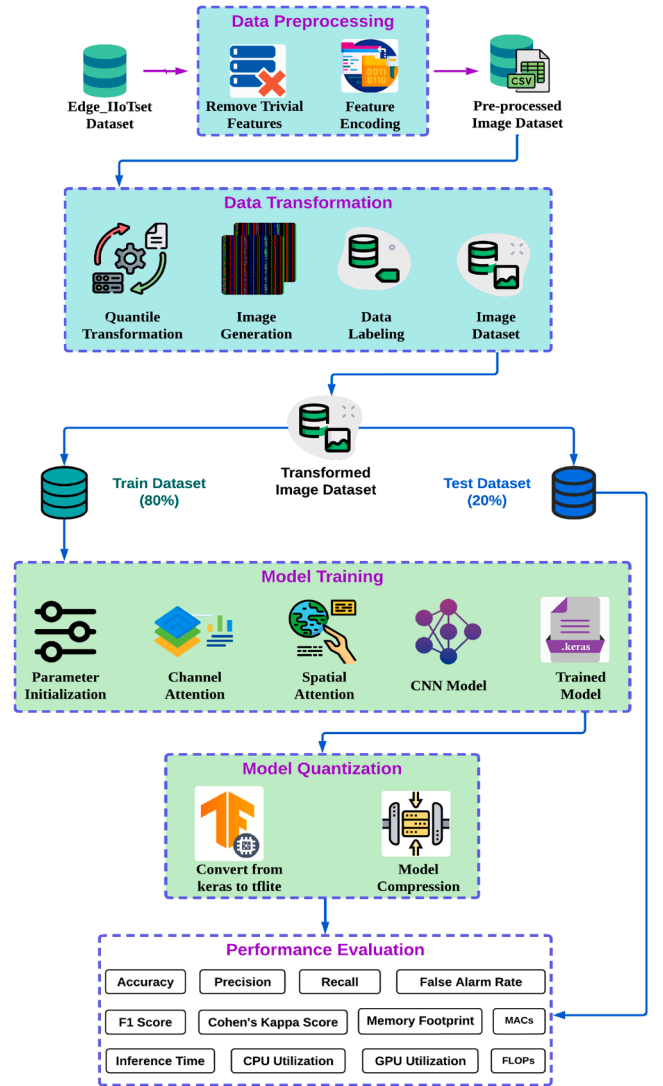


Fig. 1. Block diagram depicting the overall workflow of the proposed framework, including key components and process flow.

Moreover, Edge\_IIoTset is generated using a realistic testbed environment that includes heterogeneous edge devices, such as Raspberry Pi and Arduino boards, and collects traffic across multiple communication protocols, making it highly representative of modern industrial IoT systems.

Compared to other datasets, such as NSL-KDD or CIC-IDS2017, Edge\_IIoTset offers more contextual and device-level diversity, making it particularly suitable for evaluating attention-based models intended for deployment in resource-constrained and protocol-rich IoT/IIoT environments.

This selection ensures that the proposed model is not only tested under controlled conditions but also reflects the complexity of real-world deployment scenarios, thereby enhancing the generalizability of our approach.

#### 3.2. Data pre-processing and transformation

To prepare the Edge\_IIoTset dataset for machine learning, we initially verified that no missing values (NaNs) were present and removed 815 duplicate rows, accounting for only 0.037 % of the dataset. We then refined the dataset by discarding irrelevant features, such as network protocol details (e.g., "arp.src.proto\_ipv4"), payload data

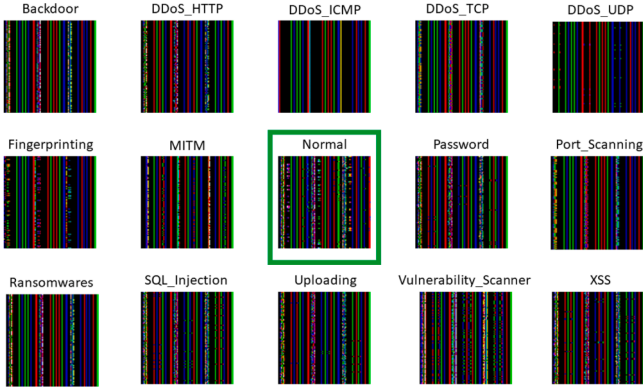


Fig. 2. Representative samples from each of the 14 distinct cyberattack categories and normal data, highlighting visual differences across attack classes.

Table 1

Overview of the pre-processed Edge\_IIoTset dataset with Key features and statistics.

| Dataset | Records   | Features | Classes | Transformed images | Feature types    |
|---------|-----------|----------|---------|--------------------|------------------|
| Train   | 1,527,737 | 96       | 15      | 5369               | Numerical (47)   |
| Test    | 381,934   | 96       | 15      | 1342               | Categorical (49) |
| Total   | 1,909,671 | 96       | 15      | 6711               | 96               |

(e.g., “http.file\_data”), and timestamps (e.g., “frame.time”), resulting in a reduced set of 47 features across 1,909,671 rows. Of these, eight columns contained categorical data, and dummy encoding was applied to seven columns excluding the target variable, “Attack\_type.” This encoding expanded the dataset to 96 features, which were subsequently saved as a CSV file for additional investigation.

To utilize pre-trained CNN architectures designed for image data, the tabular Edge\_IIoTset dataset was transformed into an image-compatible format. Quantile normalization was applied to scale feature values to a range (0–255), mimicking pixel intensities while mitigating the impact of outliers by centering values near the median. Samples were then organized into image blocks: every 285 consecutive samples, each with 95 features, formed a 95x95x3 color image. These images were labeled based on attack patterns, with ‘Normal’ assigned to blocks containing only normal samples, and the most frequent attack type assigned to mixed blocks. The labels covered 14 attack categories, including DDoS, scanning, password attacks, backdoors, XSS, and ransomware, offering a visual basis for classification through CNNs. Fig. 2 presents representative images for each of these attack categories, visually illustrating the transformed data for CNN processing. Table 1 presents the summary of the dataset after preprocessing and transformation.

### 3.3. Attention mechanism in CNNs

In deep learning, attention mechanisms are pivotal for enhancing a model’s ability to focus on the most relevant features within an input. Channel and spatial attention are two primary types of attention mechanisms used in CNNs to refine feature extraction, allowing the model to prioritize important aspects of the data for specific tasks such as image classification or attack detection.

#### 3.3.1. Channel attention

Channel attention mechanism assigns different levels of importance to each feature channel within a neural network, enabling models to prioritize the most informative features. In CNNs, each channel represents distinct learned features, such as edges or textures. Channel attention allows the network to emphasize these critical channels while reducing the influence of less relevant features, improving classification accuracy. Common implementations like Squeeze-and-Excitation (SE) blocks and

CBAM are widely used for channel-wise recalibration (Hu et al., 2018; Woo et al., 2018). To summarize, the channel attention mechanism highlights informative channels in the feature map.

Suppose the input feature map  $\mathbf{X}$  has dimensions  $H \times W \times C$ , where  $H$  and  $W$  are the spatial dimensions (height and width), and  $C$  is the number of feature channels. The goal is to enhance the feature map by emphasizing important channels using a channel-wise attention mechanism. The following steps are performed:

**1. Global Average Pooling (GAP):** In this step, we compute the average value for each channel across all spatial locations:

$$\mathbf{F}_{\text{avg}} = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W \mathbf{X}(i, j, c), \quad \forall c \in [1, C]. \quad (1)$$

The result is a vector  $\mathbf{F}_{\text{avg}} \in \mathbb{R}^C$ , where each element represents the average feature intensity for a corresponding channel. This captures global contextual information for each channel.

**2. Global Max Pooling (GMP):** Next, we compute the maximum value for each channel across all spatial locations:

$$\mathbf{F}_{\text{max}} = \max_{i,j} \mathbf{X}(i, j, c), \quad \forall c \in [1, C]. \quad (2)$$

The result is another vector  $\mathbf{F}_{\text{max}} \in \mathbb{R}^C$ , where each element represents the most prominent feature intensity for the corresponding channel. This captures the strongest activations for each channel.

**3. Shared Fully Connected Layers:** The output vectors from GAP and GMP are processed through two fully connected layers to model channel-wise dependencies:

- **First Dense Layer:** The vectors  $\mathbf{F}_{\text{avg}}$  and  $\mathbf{F}_{\text{max}}$  are passed through a dense layer with weight matrix  $\mathbf{W}_1$ , which reduces their dimensionality by a factor of  $r$ . A ReLU activation function is applied to introduce non-linearity:

$$\mathbf{F}'_{\text{avg}} = \text{ReLU}(\mathbf{W}_1 \mathbf{F}_{\text{avg}}) \quad (3)$$

$$\mathbf{F}'_{\text{max}} = \text{ReLU}(\mathbf{W}_1 \mathbf{F}_{\text{max}}) \quad (4)$$

The result is two compressed representations of the channel-wise features.

- **Second Dense Layer:** The compressed vectors are passed through another dense layer with weight matrix  $\mathbf{W}_2$ , which restores their original dimensionality:

$$\mathbf{F}''_{\text{avg}} = \mathbf{W}_2 \mathbf{F}'_{\text{avg}} \quad (5)$$

$$\mathbf{F}''_{\text{max}} = \mathbf{W}_2 \mathbf{F}'_{\text{max}}. \quad (6)$$

**4. Attention Aggregation:** The two processed vectors are combined using element-wise addition to aggregate the attention information:

$$\mathbf{F}_{\text{attn}} = \mathbf{F}''_{\text{avg}} + \mathbf{F}''_{\text{max}}. \quad (7)$$

This step integrates both average and maximum pooling results into a unified attention representation.

**5. Channel Attention Weights:** To generate attention weights for each channel, a sigmoid activation function is applied to  $\mathbf{F}_{\text{attn}}$ :

$$\mathbf{A}_{\text{channel}} = \sigma(\mathbf{F}_{\text{attn}}), \quad (8)$$

where  $\mathbf{A}_{\text{channel}} \in \mathbb{R}^C$  is a vector of normalized weights (between 0 and 1) for each channel. These weights indicate the importance of each channel.

**6. Final Attention-Weighted Output:** The attention weights are used to scale the input feature map channel-wise. For each spatial position  $(i, j)$  and channel  $c$ , the output feature map is computed as:

$$\mathbf{X}_{\text{channel-out}}(i, j, c) = \mathbf{X}(i, j, c) \cdot \mathbf{A}_{\text{channel}}(c). \quad (9)$$

This step enhances the input feature map by emphasizing channels with higher attention weights while suppressing less important ones.

By following these steps, the mechanism effectively focuses on the most informative channels, improving the model's ability to capture critical patterns in the input data.

### 3.3.2. Spatial attention

Spatial attention focuses on identifying and emphasizing the most relevant regions within an input feature map. Rather than focusing on entire channels, spatial attention highlights spatial areas (height and width) that are informative for the task at hand. For tasks like attack detection, spatial attention helps the model concentrate on suspicious patterns in specific regions, improving accuracy by downplaying less relevant areas. Combined with channel attention, spatial attention forms a powerful approach to refine feature maps in high-dimensional data. To summarize, the spatial attention mechanism highlights informative spatial regions.

*Compute a single-channel spatial descriptor.* To generate spatial attention weights, the mechanism first aggregates information across all channels of the input feature map  $\mathbf{X}$ . This is achieved by applying a  $k \times k$  convolutional filter, where  $k$  defines the kernel size, capturing spatial context from a local neighborhood. In this case,  $k = 7$  is used to capture a wide receptive field.

The result of the convolution is passed through a sigmoid activation function ( $\sigma$ ) to normalize the spatial weights between 0 and 1:

$$\mathbf{A}_{\text{spatial}} = \sigma(\text{Conv}_{k \times k}(\mathbf{X})) \quad (10)$$

Here:

- $\text{Conv}_{k \times k}(\mathbf{X})$  denotes the convolution operation using a  $k \times k$  kernel applied to the input feature map  $\mathbf{X}$ .
- $\sigma$  is the sigmoid activation function, ensuring the spatial attention weights are scaled within a range of  $[0, 1]$ .

The result,  $\mathbf{A}_{\text{spatial}} \in \mathbb{R}^{H \times W \times 1}$ , is a single-channel spatial descriptor that assigns an attention weight to each spatial position  $(i, j)$  in the feature map. These weights indicate the importance of individual spatial regions.

*Compute the final attention-weighted output.* Once the spatial attention weights  $\mathbf{A}_{\text{spatial}}$  are generated, they are used to modulate the original input feature map  $\mathbf{X}$ . Since  $\mathbf{A}_{\text{spatial}}$  has only one channel ( $\mathbb{R}^{H \times W \times 1}$ ), it is broadcasted across all  $C$  channels of  $\mathbf{X}$ . For each spatial position  $(i, j)$  and channel  $c$ , the attention mechanism enhances or suppresses the feature map values by scaling them with the corresponding spatial attention weight:

$$\mathbf{X}_{\text{spatial-out}}(i, j, c) = \mathbf{X}(i, j, c) \cdot \mathbf{A}_{\text{spatial}}(i, j) \quad (11)$$

Here:

- $\mathbf{X}(i, j, c)$  is the value of the input feature map at spatial position  $(i, j)$  and channel  $c$ .
- $\mathbf{A}_{\text{spatial}}(i, j)$  is the spatial attention weight for the position  $(i, j)$ , shared across all channels.
- $\mathbf{X}_{\text{spatial-out}}$  represents the final feature map with spatial attention applied, having the same dimensions as the input  $\mathbf{X}$ :  $H \times W \times C$ .

This operation ensures that important spatial regions are highlighted across all channels, while less significant regions are suppressed.

By integrating channel attention (which selects important feature channels) and spatial attention (which emphasizes critical spatial regions), the model can learn a comprehensive attention mechanism that considers both inter-channel and intra-spatial dependencies. This dual attention approach significantly enhances the model's ability to focus on the most informative features, leading to improved performance in a variety of tasks such as image classification, object detection, and semantic segmentation. An illustration of channel and attention mechanisms is presented in Fig. 3. We provide a detailed description of Fig. 3 in the following paragraphs.

*Channel attention module.* The input feature map  $\mathbf{X}$  is first processed using *Global Average Pooling* and *Global Max Pooling* to produce  $F_{\text{avg}}$  and  $F_{\text{max}}$ , respectively. These pooled features are then passed through a shared multi-layer perceptron (MLP) composed of two fully connected layers (FC1 and FC2), yielding two intermediate outputs  $F'_{\text{avg}}$  and  $F'_{\text{max}}$ . These outputs are combined using element-wise addition to form the aggregated attention feature  $F_{\text{attn}}$ , which is subsequently passed through a sigmoid activation function  $\sigma$  to generate the channel attention map  $\mathbf{A}_{\text{channel}}$ . Finally, this attention map is multiplied element-wise with the original input  $\mathbf{X}$  to obtain the channel-refined feature map.

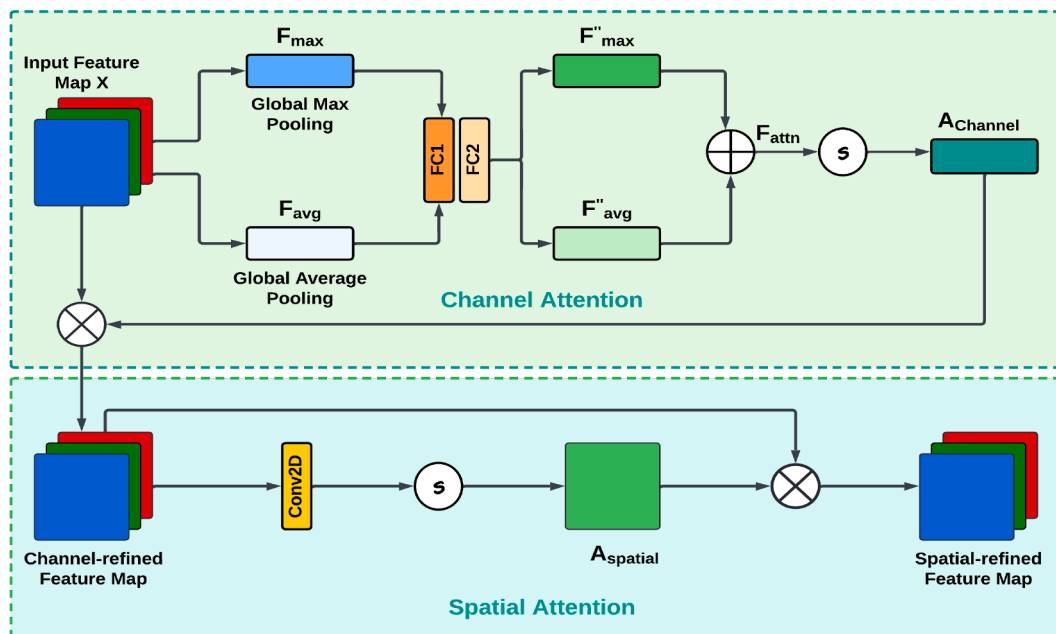


Fig. 3. Diagram illustrating the functionality of channel and spatial attention mechanisms for enhanced feature extraction.

**Spatial attention module.** The channel-refined feature map is then passed through a 2D convolutional layer (Conv2D) to extract spatial context. The output is activated using a sigmoid function to produce the spatial attention map  $A_{\text{spatial}}$ . This map is multiplied element-wise with the channel-refined feature map to obtain the final spatial-refined feature map, which is fed into the subsequent layers of the CNN.

This dual-attention mechanism enables the model to adaptively focus on both significant channels and spatial regions within the feature maps, thereby enhancing feature representation and improving intrusion detection performance.

### 3.4. Deep transfer learning

Transfer learning allows models to leverage pre-trained weights from large datasets, such as ImageNet, which contains over a million labeled images across numerous categories. These pre-trained weights provide a foundation of general feature extraction capabilities that can be fine-tuned for specific tasks like IoT attack detection. This approach reduces training time, requires less data, and often improves accuracy by starting from a model that has already learned basic image features (Deng et al., 2009).

### 3.5. ResNet50

ResNet50, a 50-layer CNN introduced by He et al. (2016), addresses the vanishing gradient problem through residual connections, also known as “skip connections.” These connections allow gradients to flow through deeper layers without vanishing, making it possible for the network to learn complex hierarchical features. This architecture has achieved high performance on a variety of image tasks and is effective for transfer learning due to its robust feature extraction capabilities. When combined with attention mechanisms, ResNet50 can further enhance detection accuracy by focusing on relevant data features (He et al., 2016).

### 3.6. MobileNet

MobileNet, developed by Howard et al., is a CNN optimized for efficient, real-time applications on mobile and IoT devices. By using depth-wise separable convolutions, MobileNet reduces the number of parameters and computational costs, making it ideal for resource-limited environments. MobileNet models pre-trained on ImageNet weights can be fine-tuned for specific applications, and when augmented with attention mechanisms, they can selectively focus on important features and regions, making them suitable for attack detection tasks in IoT (Howard, 2017).

### 3.7. Model quantization

Quantization of neural networks is a sophisticated compression technique that reduces model resource consumption while preserving performance. This process involves transforming the continuous, high-precision values of a neural network’s weights and activations into lower-bit, discrete representations. By decreasing precision, quantization effectively reduces model size and computational demands, making it well-suited for deployment on resource-limited platforms such as mobile devices and edge computing environments. Quantized models require less memory and enable faster inference, which enhances efficiency and reduces latency (Krishnamoorthi, 2018). Additionally, by introducing noise during training, quantization can act as a form of regularization, potentially helping to mitigate overfitting (Hubara et al., 2018). This makes quantization a valuable technique for optimizing deep learning models in scenarios where performance and efficiency must be balanced.

## 4. Implementation setup and evaluation

This section outlines the implementation details of the proposed framework and the metrics utilized for evaluating its performance.

### 4.1. Implementation setup

We implemented the proposed Deep Transfer Learning (DTL) architecture and tested its performance on Google Colab Pro, utilizing the Keras library. The platform provides an Nvidia Turing T4 GPU with 25 GB of GPU memory and 15 GB of RAM, ensuring efficient execution of deep learning algorithms. To start, we used Keras’s ImageDataGenerator to load the training and testing image datasets directly from Google Drive. Next, we integrated pre-trained CNN models for ResNet50 and MobileNet, using ImageNet weights, into the Keras environment. During training, we applied a callback function to store the model achieving the highest validation accuracy, setting the number of epochs to 10. The models that achieved the highest performance were saved to Google Drive for future reference, and their effectiveness was then assessed using the testing dataset.

### 4.2. Performance evaluation metrics

To ensure a shared foundation for evaluating performance metrics, it is crucial to begin by reviewing and clarifying some key concepts. This step establishes a common understanding, allowing for a more consistent and accurate interpretation of the evaluation criteria.

**True Positive (TP):** This occurs when the model correctly identifies a positive instance. For example, in a medical diagnosis system, the model accurately detects a patient with a specific disease.

**True Negative (TN):** This happens when the model correctly identifies a negative instance. For instance, a spam filter correctly identifies a legitimate email as not being spam.

**False Positive (FP):** This occurs when the model incorrectly labels a negative instance as positive. An example would be a facial recognition system incorrectly identifying a person as someone else on the watchlist.

**False Negative (FN):** This happens when the model fails to identify a positive instance and labels it as negative. For example, a fraud detection system overlooks a fraudulent transaction, marking it as legitimate.

The effectiveness of the proposed scheme is assessed using the following performance metrics.

- Accuracy

The ratio of correctly classified instances to the total number of instances.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (12)$$

- Precision

The proportion of true positive predictions out of all positive predictions.

$$Precision = \frac{TP}{TP + FP} \quad (13)$$

- Recall

The proportion of true positive cases that were correctly identified by the model.

$$Recall = \frac{TP}{TP + FN} \quad (14)$$

- False Alarm Rate (FAR)

The ratio of false positives to all actual negatives, representing the rate of incorrect positive predictions.

$$FAR = \frac{FP}{FP + TN} \quad (15)$$

- F1 Score

The harmonic mean of precision and recall, providing a single metric for model performance on positive cases.

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (16)$$

- Cohen's Kappa Score (CKS)

A measure of agreement between predicted and actual classifications, adjusted for the probability of random agreement.

$$CKS = \frac{\text{Observed Accuracy} - \text{Expected Accuracy}}{1 - \text{Expected Accuracy}} \quad (17)$$

where

*Observed Accuracy* refers to the proportion of instances where raters agree when assigning items to specific categories.

*Expected Accuracy* represents the degree of agreement that would occur purely by chance.

- Memory footprint

The total memory required by a model during execution, essential for deployment on memory-constrained devices.

- Inference time

The time taken by a trained model to process a single input and produce an output, typically measured in milliseconds (ms) or seconds (s).

- CPU Utilization

The percentage of CPU capacity used by a model during operation, reflecting the efficiency of CPU resource use.

- GPU Utilization

The percentage of GPU capacity utilized by a model, providing insight into how effectively the model leverages GPU resources.

- Floating Point Operations Per Second (FLOPs)

A measure of computational efficiency, indicating the number of floating-point operations a model performs per second.

- Multiply-Accumulate Operations (MACs)

The total number of multiplications and additions a model performs during a forward pass, impacting computational complexity.

## 5. Experimental results

The experimentation and performance analysis were conducted in two main stages. In the first stage, a set of random hyperparameters was assigned to each base model, and each model was trained using these parameters. The models were successfully saved after training. In the second stage, channel and spatial attention mechanisms were applied to the input data, and the models were trained again using the same set of hyperparameters. These attention-augmented models were also saved after training.

To ensure consistency in the results, the number of epochs was fixed at 10 for both stages. During each epoch, the model with the highest validation accuracy was saved to retain the highest-quality version of each model. The final selection of top-performing base models was made by comparing validation accuracy across all epochs, with the highest-accuracy models chosen for further use. All trained models were securely

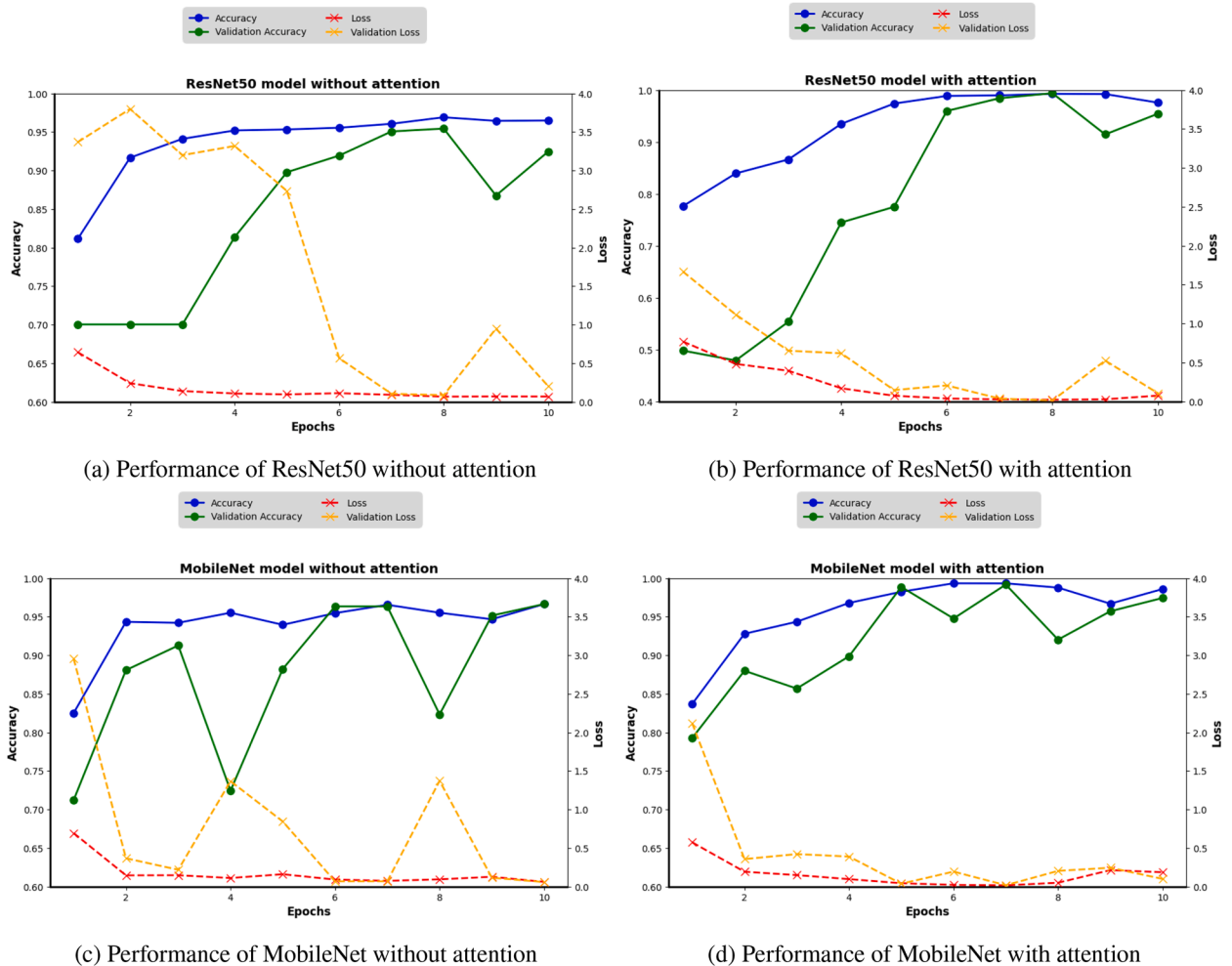


Fig. 4. Comparison of Training and Testing accuracies and losses for ResNet50 and MobileNet models, highlighting performance improvements achieved with the integration of attention mechanisms.

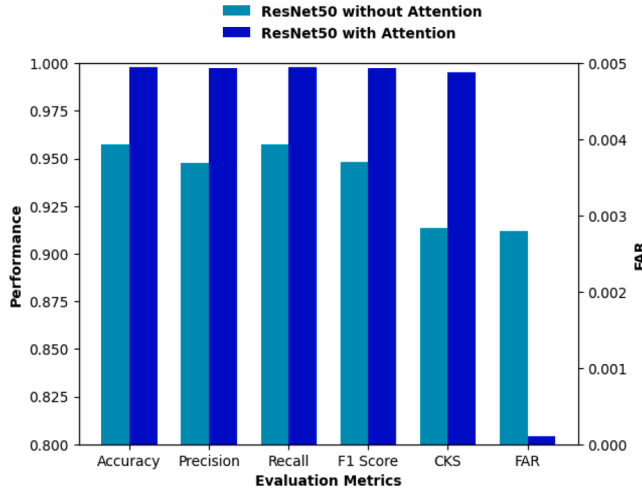


Fig. 5. Comprehensive evaluation of ResNet50 Model's overall performance across key metrics.

stored in Google Drive to enable future use for predictions. The train and test accuracies as well as train and test losses for ResNet50 and MobileNet models, before and after attention, are presented in Fig. 4. The attention-augmented models demonstrate improved accuracy and reduced loss than the models without attention. We can note that the overall performance of both models in case of attention is better.

### 5.1. Results

We load the trained models from Google Drive and perform evaluations for accuracy, precision, recall, false alarm rate, F1 score and cohen's kappa score. Figs. 5 and 6 show the overall performance of ResNet50 and MobileNet models respectively. The performance of FAR is shown on the secondary axis, because of its small values, whereas the performance of other metrics has been shown on the primary axis. We can see that the attention-augmented models perform better than models without attention. Table 2 summarizes the performance results of both models before and after attention.

The Edge-IIoTset dataset comprises a total of 1,909,671 samples, including both benign and attack instances. Among these, 1,363,998 are benign samples and 545,673 are attack samples, distributed across 15

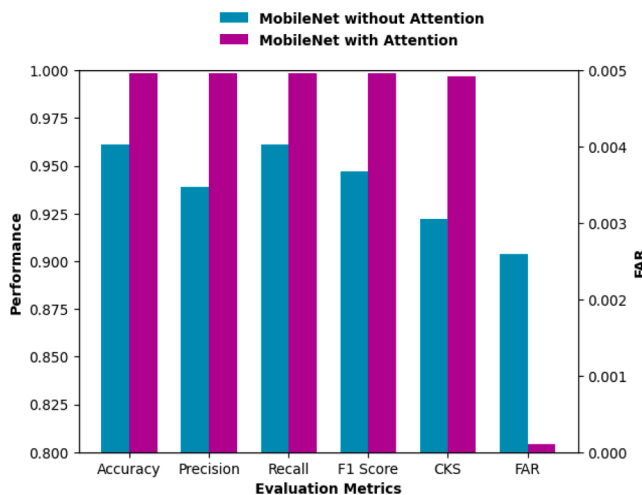


Fig. 6. Comprehensive evaluation of MobileNet Model's overall performance across key metrics.

Table 2

Performance comparison of ResNet50 and MobileNet models before and after integrating attention mechanisms across key metrics.

| Model     | Attention | Accuracy | Precision | Recall | FAR    | F1 Score | CKS    |
|-----------|-----------|----------|-----------|--------|--------|----------|--------|
| ResNet50  | No        | 0.9574   | 0.9475    | 0.9574 | 0.0028 | 0.9482   | 0.9136 |
| ResNet50  | Yes       | 0.9977   | 0.9976    | 0.9977 | 0.0001 | 0.9976   | 0.9954 |
| MobileNet | No        | 0.9612   | 0.9389    | 0.9612 | 0.0026 | 0.9472   | 0.9222 |
| MobileNet | Yes       | 0.9985   | 0.9986    | 0.9985 | 0.0001 | 0.9983   | 0.9970 |

Table 3

Performance evaluation for ResNet50 without attention mechanism across all individual attack classes.

| Attack class          | Accuracy | Precision | Recall | FAR    | F1 Score |
|-----------------------|----------|-----------|--------|--------|----------|
| Backdoor              | 0.5300   | 1.0000    | 0.5300 | 0.0000 | 0.6928   |
| DDoS_HTTP             | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| DDoS_ICMP             | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| DDoS_TCP              | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| DDoS_UDP              | 1.0000   | 0.9804    | 1.0000 | 0.0014 | 0.9901   |
| Fingerprinting        | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| MITM                  | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| Normal                | 1.0000   | 0.5714    | 1.0000 | 0.0536 | 0.7273   |
| Password              | 0.9800   | 0.4279    | 0.9800 | 0.0936 | 0.5957   |
| Port_Scanning         | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| Ransomware            | 0.4300   | 0.5972    | 0.4300 | 0.0207 | 0.5000   |
| SQL_Injection         | 0.0000   | 0.0000    | 0.0000 | 1.0000 | 0.0000   |
| Uploading             | 0.6900   | 1.0000    | 0.6900 | 0.0000 | 0.8166   |
| Vulnerability_Scanner | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| XSS                   | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |

attack classes. The dataset exhibits a noticeable imbalance, with certain attack types being underrepresented compared to others.

To mitigate the effects of this imbalance, we employed multiple evaluation metrics beyond accuracy, including precision, recall, F1-score, and Cohen's Kappa, which are better suited for imbalanced data. These metrics provide a more comprehensive and reliable assessment of our model's performance, especially in detecting minority classes.

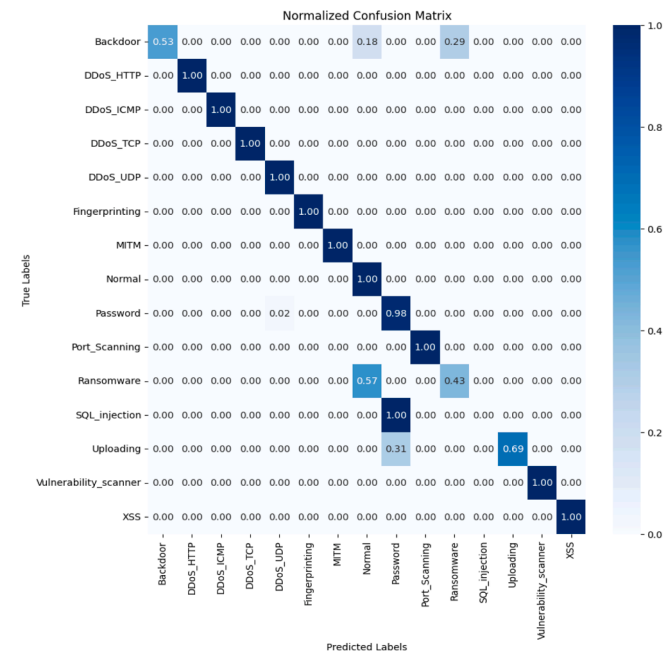
To further delve into the attack detection of these models, we analyze the performance of each attack class separately, before and after the application of attention. For the ResNet50 model, Table 3 summarizes the performance results of all attack classes before attention. Backdoor, Password, Ransomware and Uploading attack classes have accuracies of 53 %, 98 %, 43 % and 69 %, respectively. SQL\_Injection class has been completely misclassified. All other classes have been accurately classified. Now we evaluate the attention-augmented ResNet50 model. Table 4 shows performance results of all attack classes after attention. Backdoor, Password and Ransomware attack classes have accuracy of 94 %, 98 % and 57 %, respectively. All other classes have been accurately labeled. Fig. 7 shows the results in the form of a normalized confusion matrix before attention, whereas Fig. 8 shows the normalized confusion matrix for attack detection after attention. We can clearly see from the comparison of these results that the attack detection performance of ResNet50 has improved after the application of attention. Moreover, complex classes, which were misclassified in the attention-less version, have now been accurately classified or the attack detection accuracy has improved.

For MobileNet model, Table 5 summarizes the performance results of all individual attack classes before attention. Ransomware attack class has an accuracy of 14 % whereas Fingerprinting and SQL\_Injection classes have been completely misclassified. All other classes have been accurately classified. Now we evaluate the attention-augmented MobileNet model. Table 6 shows performance results of all attack classes after attention. Attack detection accuracy of Ransomware class has been improved to 43 % whereas all other classes have been accurately labeled. Furthermore, Fig. 9 shows the results in the form of a normalized confusion matrix before attention, whereas Fig. 10 shows normalized confusion matrix for attack detection after attention. We can conclude

**Table 4**

Performance evaluation for ResNet50 with attention mechanism across all individual attack classes.

| Attack class          | Accuracy | Precision | Recall | FAR    | F1 Score |
|-----------------------|----------|-----------|--------|--------|----------|
| Backdoor              | 0.9400   | 0.7642    | 0.9400 | 0.0207 | 0.8430   |
| DDoS_HTTP             | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| DDoS_ICMP             | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| DDoS_TCP              | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| DDoS_UDP              | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| Fingerprinting        | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| MITM                  | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| Normal                | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| Password              | 0.9800   | 1.0000    | 0.9800 | 0.0000 | 0.9899   |
| Port_Scanning         | 1.0000   | 0.8772    | 1.0000 | 0.0100 | 0.9346   |
| Ransomware            | 0.5700   | 0.9048    | 0.5700 | 0.0043 | 0.6994   |
| SQL_Injection         | 1.0000   | 0.9804    | 1.0000 | 0.0014 | 0.9901   |
| Uploading             | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| Vulnerability_Scanner | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| XSS                   | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |

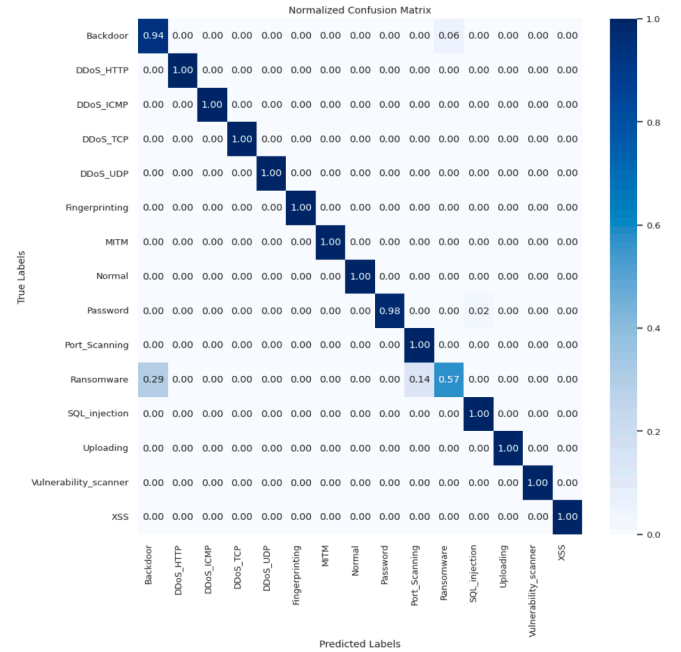


**Fig. 7.** Confusion Matrix for ResNet50 without attention mechanisms: This figure displays the model's classification performance on the Edge-IIoT dataset, highlighting the true and false predictions across 14 attack classes. Diagonal elements represent correct classifications, while off-diagonal elements indicate misclassifications.

from the comparison of these results that MobileNet has an improved attack detection performance when attention is applied. Moreover, complex classes which were misclassified in attentionless version have now been accurately classified or attack detection accuracy has improved.

## 5.2. Impact of model quantization

TensorFlow Lite offers an efficient conversion process designed to transform trained Keras models into a more compact and optimized format. This process is especially beneficial for deploying machine learning models on devices with limited resources. The conversion begins with the TensorFlow Lite Converter, which produces a standard TFLite model while maintaining the primary functionality of the original Keras model, though without any extra optimizations. Following this, we apply post-training quantization—a technique that further reduces model size and increases inference speed by representing weights and



**Fig. 8.** Confusion Matrix for ResNet50 with Attention Mechanisms: This figure displays the model's classification performance on the Edge-IIoT dataset, highlighting the true and false predictions across 14 attack classes. Diagonal elements represent correct classifications, while off-diagonal elements indicate misclassifications.

**Table 5**

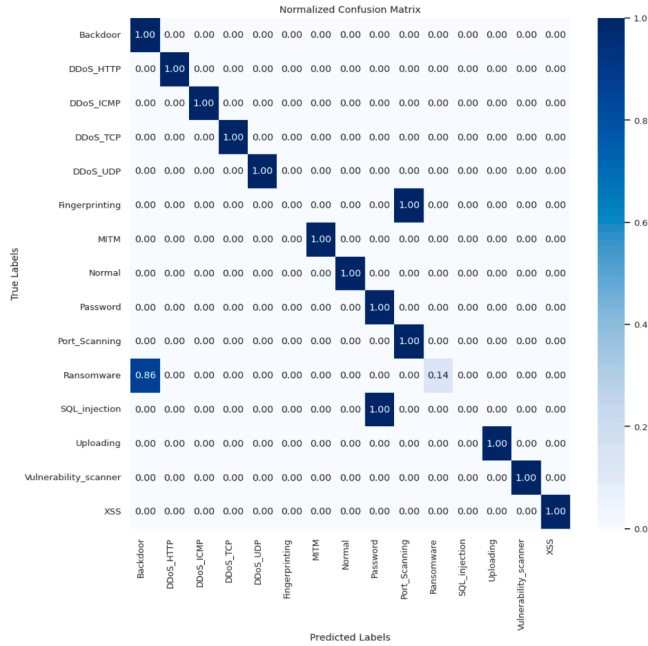
Performance evaluation for MobileNet without attention mechanism across all individual attack classes.

| Attack class          | Accuracy | Precision | Recall | FAR    | F1 Score |
|-----------------------|----------|-----------|--------|--------|----------|
| Backdoor              | 1.0000   | 0.5376    | 1.0000 | 0.0614 | 0.6993   |
| DDoS_HTTP             | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| DDoS_ICMP             | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| DDoS_TCP              | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| DDoS_UDP              | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| Fingerprinting        | 0.0000   | 0.0000    | 0.0000 | 1.0000 | 0.0000   |
| MITM                  | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| Normal                | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| Password              | 1.0000   | 0.5000    | 1.0000 | 0.0714 | 0.6667   |
| Port_Scanning         | 1.0000   | 0.5000    | 1.0000 | 0.0714 | 0.6667   |
| Ransomware            | 0.1400   | 1.0000    | 0.1400 | 0.0000 | 0.2456   |
| SQL_Injection         | 0.0000   | 0.0000    | 0.0000 | 1.0000 | 0.0000   |
| Uploading             | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| Vulnerability_Scanner | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| XSS                   | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |

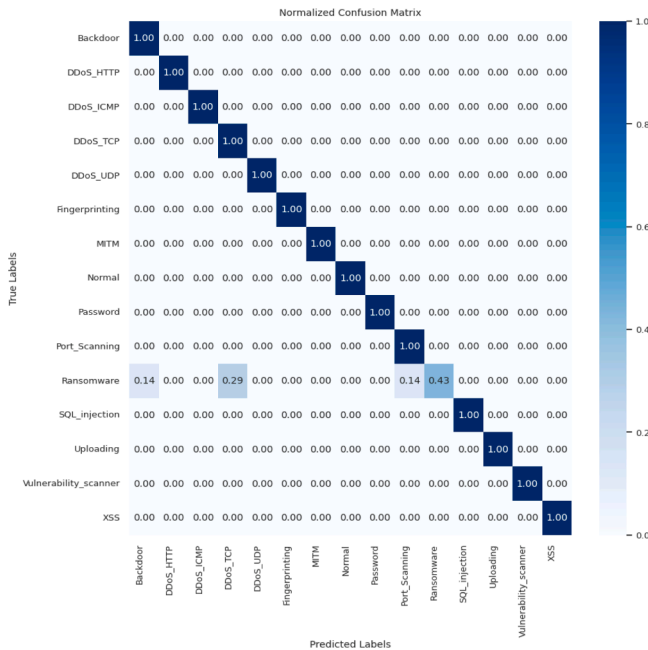
**Table 6**

Performance evaluation for MobileNet with attention mechanism across all individual attack classes.

| Attack class          | Accuracy | Precision | Recall | FAR    | F1 Score |
|-----------------------|----------|-----------|--------|--------|----------|
| Backdoor              | 1.0000   | 0.8772    | 1.0000 | 0.0100 | 0.9346   |
| DDoS_HTTP             | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| DDoS_ICMP             | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| DDoS_TCP              | 1.0000   | 0.7752    | 1.0000 | 0.0207 | 0.8734   |
| DDoS_UDP              | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| Fingerprinting        | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| MITM                  | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| Normal                | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| Password              | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| Port_Scanning         | 1.0000   | 0.8772    | 1.0000 | 0.0100 | 0.9346   |
| Ransomware            | 0.4300   | 1.0000    | 0.4300 | 0.0000 | 0.6014   |
| SQL_Injection         | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| Uploading             | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| Vulnerability_Scanner | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |
| XSS                   | 1.0000   | 1.0000    | 1.0000 | 0.0000 | 1.0000   |



**Fig. 9.** Confusion Matrix for MobileNet without Attention Mechanisms: This figure displays the model's classification performance on the Edge-IIoT dataset, highlighting the true and false predictions across 14 attack classes. Diagonal elements represent correct classifications, while off-diagonal elements indicate misclassifications.



**Fig. 10.** Confusion Matrix for MobileNet with Attention Mechanisms: This figure displays the model's classification performance on the Edge-IIoT dataset, highlighting the true and false predictions across 14 attack classes. Diagonal elements represent correct classifications, while off-diagonal elements indicate misclassifications.

activations with lower precision. This quantized version offers a highly compressed model structure optimized for edge devices and systems with limited computational power, achieving increased efficiency with minimal impact on model performance.

To further evaluate resource utilization, we applied quantization to the selected models. The trained Keras models were first converted to

TFLite format, followed by default optimization and quantization to further compress their size. We do a comparative analysis of memory footprints for the quantized models, showing substantial reductions in model size. These quantized models are then assessed for inference time, CPU and GPU utilization, and computational efficiency and complexity. All the results for the mentioned evaluations are summarized in Table 9. The results show that the size has been reduced by almost four times; however, a slight increase in almost all metrics is noted when attention is applied. Attention-based CNN models generally have a slightly higher computational overhead compared to standard CNN models. This additional overhead arises because attention mechanisms introduce extra operations, such as pooling, weighting, and additional matrix multiplications, to recalibrate feature maps based on channel and spatial importance. These operations increase the number of parameters and computations, resulting in a modest increase in model size, memory usage, and inference time.

However, despite this slight increase in overhead, attention mechanisms often lead to significantly better performance in terms of accuracy and feature extraction, especially for tasks requiring high sensitivity to specific patterns, such as object detection, attack detection, and cybersecurity. Therefore, the trade-off between improved model performance and the additional computational cost is often justified, particularly in scenarios where accuracy is prioritized.

To summarize, attention-based CNN models do have a slightly higher overhead, but the impact is generally modest and is outweighed by the improvements in model accuracy and interpretability. This makes attention-based models a valuable choice for applications where high performance is critical.

### 5.3. Ablation study on attention mechanisms

To evaluate the individual contributions of channel and spatial attention mechanisms, we conduct an ablation study using the MobileNet and ResNet50 architectures. We integrate each attention type separately and compare it against the baseline and combined attention model. The results reveal that channel attention consistently outperforms spatial attention, particularly on the transformed image dataset. The performance comparisons of the ablation study are presented in the Table 7.

### 5.4. Cross-dataset validation and comparison with LSTM

To further evaluate the robustness and generalizability of the proposed approach with channel and spatial attention, additional experiments were conducted on the CIC-IDS2017 dataset using the MobileNet model. The attention-enhanced MobileNet achieved 99.78 % accuracy, significantly outperforming its non-attention counterpart, which achieved 97.38 % accuracy. These results confirm that integrating both channel and spatial attention mechanisms enhances the model's ability to detect attacks, even when applied to a different dataset with varying

**Table 7**

Ablation study: Comparison of classification performance using different attention configurations.

| Model     | Attention mechanism      | Accuracy      | Precision     | Recall        | F1 Score      | CKS           |
|-----------|--------------------------|---------------|---------------|---------------|---------------|---------------|
| ResNet50  | No Attention             | 0.9574        | 0.9475        | 0.9574        | 0.9482        | 0.9136        |
|           | Channel Attention        | 0.9761        | 0.9803        | 0.9761        | 0.9760        | 0.9519        |
|           | Spatial Attention        | 0.9663        | 0.9718        | 0.9663        | 0.9650        | 0.9321        |
|           | <b>Channel + Spatial</b> | <b>0.9977</b> | <b>0.9976</b> | <b>0.9977</b> | <b>0.9976</b> | <b>0.9954</b> |
| MobileNet | No Attention             | 0.9612        | 0.9389        | 0.9612        | 0.9472        | 0.9222        |
|           | Channel Attention        | 0.9805        | 0.9879        | 0.9805        | 0.9829        | 0.9609        |
|           | Spatial Attention        | 0.9760        | 0.9844        | 0.9760        | 0.9754        | 0.9517        |
|           | <b>Channel + Spatial</b> | <b>0.9985</b> | <b>0.9986</b> | <b>0.9985</b> | <b>0.9983</b> | <b>0.9970</b> |

**Table 8**

Cross-dataset performance comparison of attention enhanced MobileNet with LSTM.

| Dataset      | Approach                                       | Accuracy      | Precision      | Recall        | F1 Score      | CKS           |
|--------------|------------------------------------------------|---------------|----------------|---------------|---------------|---------------|
| Edge_IIoTset | LSTM                                           | 0.9479        | 0.9595         | 0.9479        | 0.9434        | 0.8915        |
| Edge_IIoTset | <b>MobileNet + Channel + Spatial Attention</b> | <b>0.9985</b> | <b>0.99986</b> | <b>0.9985</b> | <b>0.9983</b> | <b>0.9970</b> |
| CIC-IDS2017  | MobileNet + No Attention                       | 0.9738        | 0.9769         | 0.9738        | 0.9736        | 0.9466        |
| CIC-IDS2017  | <b>MobileNet + Channel + Spatial Attention</b> | <b>0.9978</b> | <b>0.9981</b>  | <b>0.9978</b> | <b>0.9978</b> | <b>0.9955</b> |

**Table 9**

Resource comparison of ResNet50 and MobileNet models with and without attention mechanisms: This figure compares the model size, memory usage, resource utilization, computational efficiency and complexity of ResNet50 and MobileNet before and after adding attention mechanisms.

| Evaluation metric                 | ResNet50<br>without attention | ResNet50<br>with attention | MobileNet<br>without attention | MobileNet<br>with attention |
|-----------------------------------|-------------------------------|----------------------------|--------------------------------|-----------------------------|
| Float Model Size (MB)             | 91.63                         | 112.03                     | 13.224                         | 18.428                      |
| Quantized Model Size (MB)         | 23.322                        | 28.488                     | 3.483                          | 4.818                       |
| Compression Ratio                 | 3.928                         | 3.932                      | 3.796                          | 3.824                       |
| Inference Time (ms)               | 4.815                         | 5.415                      | 2.993                          | 0.207                       |
| CPU Utilization (%)               | 23.7                          | 17.4                       | 15.7                           | 27.5                        |
| GPU Utilization (%)               | 7.337                         | 14.004                     | 1.504                          | 2.337                       |
| Computational Efficiency (GFLOPs) | 7.713                         | 7.733                      | 1.135                          | 1.143                       |
| Computational Complexity (GMACs)  | 3.856                         | 3.867                      | 0.568                          | 0.572                       |

attack patterns and traffic structures. Table 8 presents a comparison of results achieved on two datasets.

The CIC-IDS2017 dataset comprises 14 distinct attack classes alongside normal traffic. These attack types include: Bot, Brute Force Web, Brute Force XSS, DDoS-HOIC, DDoS-LOIC-UDP, DDoS-LOIC-HTTP, DoS-GoldenEye, DoS-Hulk, DoS-SlowHTTPTest, DoS-Slowloris, FTP-Brute-Force, Infiltration, SQL Injection, and SSH-BruteForce. This dataset provides a diverse set of modern attack scenarios, making it a widely accepted benchmark for evaluating intrusion detection systems.

In contrast, the Edge-IIoTset dataset includes a different range of attack types tailored specifically for IoT and IIoT environments. A detailed list of the attack classes in Edge-IIoTset is presented in Tables 5 and 6. These differences in attack diversity and network context make both datasets complementary for assessing the generalizability and robustness of IDS models across varied operational settings.

Additionally, a performance comparison was conducted using an LSTM-based model trained on the Edge\_IIoTset dataset. The LSTM model achieved a validation accuracy of 94.79%, indicating its capability to capture attack patterns in the data. However, it was consistently outperformed by the proposed MobileNet model enhanced with channel and spatial attention, which achieved a significantly higher accuracy of 99.85%. These results highlight that for transformed image-based representations of attack data, attention-augmented convolutional architectures like MobileNet are substantially more effective than sequential models such as LSTM. The results of the performance comparison are presented in Table 8.

### 5.5. Real-world deployment feasibility

In addition to achieving high degree of accuracy in identifying IoT/IIoT attacks, we also take into account the practical feasibility of deploying our models in real-world scenarios. Particularly, we selected MobileNet as a baseline model due to its lightweight architecture, which

is optimized for edge devices through the use of depthwise separable convolutions.

The attention mechanisms introduced in this study, channel and spatial attention, were implemented using computationally efficient techniques such as global average pooling, max pooling, and  $7 \times 7$  convolutions. These design choices guarantee that the computational and memory expenses associated with attentional enhancements are kept to a minimum.

Notably, the attention-augmented MobileNet model has a compact size of approximately 5 MB, making it deployable on platforms with limited resources such as Raspberry Pi, Jetson Nano, and ARM Cortex-A processors. This demonstrates that the proposed model is not only effective but also practical for intrusion detection in real-world IoT and IIoT scenarios.

### 5.6. Performance evaluation against existing methods

We compare the performance of our proposed approach with several related methods discussed in Section 2. Researchers have employed various techniques, including deep learning, federated learning, and transfer learning, to enhance attack detection in IoT and IIoT systems. While some studies have explored the use of attention mechanisms, to the best of our knowledge, none have combined channel and spatial attention to improve attack detection in IoT. Additionally, our approach undergoes rigorous evaluation on the latest IoT dataset, which includes 14 types of attack data alongside normal traffic, ensuring the authenticity and robustness of our results. Beyond general performance metrics, we also assess our proposed approach in terms of quantization, resource utilization, computational efficiency, and computational complexity, distinguishing our work as a comprehensive and innovative contribution to IoT and IIoT security. The performance comparison of our proposed framework with existing approaches in literature is summarized in Table 10.

**Table 10**

Comparison of our proposed framework with existing frameworks in literature.

| Study                          | Year        | Approach                                 | Dataset                    | Accuracy                    | Quantization | Resource Utilization | Computational Efficiency | Computational Complexity |
|--------------------------------|-------------|------------------------------------------|----------------------------|-----------------------------|--------------|----------------------|--------------------------|--------------------------|
| RNN-IDS (Yin et al., 2017)     | 2017        | RNN                                      | NSL-KDD                    | 97.09 %                     | No           | No                   | No                       | No                       |
| S-NDAE (Shone et al., 2018)    | 2018        | Stacked Autoencoders                     | KDD Cup '99                | 97.85 %                     | No           | No                   | No                       | No                       |
| Hussain et al. (2020)          | 2020        | TL, ResNet18                             | CICDDoS2019                | 87 %                        | No           | No                   | No                       | No                       |
| IoTDefender (Fan et al., 2020) | 2020        | FTL, CNN                                 | CICIDS2017                 | 91.93 %                     | No           | No                   | No                       | No                       |
| Guo et al. (2021)              | 2021        | Transformer                              | HDFS, BGL Thunderbird      | 82.32 %, 90.83 %<br>96.64 % | No           | No                   | No                       | No                       |
| Alkadi et al. (2020)           | 2021        | RNN-BiLSTM                               | BoT-IoT                    | 97.99 %                     | No           | No                   | No                       | No                       |
| Ge et al. (2021)               | 2021        | TL, FNN                                  | Bot-IoT                    | 99.79 %                     | No           | No                   | No                       | No                       |
| Guan et al. (2021)             | 2021        | TL                                       | USTC-TFC2016               | 98.91 %                     | No           | No                   | No                       | No                       |
| Awan et al. (2021)             | 2021        | VGG19 + Spatial attention                | Malware image dataset      | 96 %                        | No           | No                   | No                       | No                       |
| TranAD (Tuli et al., 2022)     | 2022        | Transformer                              | SWaT, WADI                 | 84.91 %, 89.68 %            | No           | No                   | No                       | No                       |
| E-GraphSAGE (Lo et al., 2022)  | 2022        | GNN                                      | NF-BoT-IoT<br>TON-IoT      | 81 %<br>87 %                | No           | No                   | No                       | No                       |
| Long et al. (2023)             | 2023        | AGCA                                     | UNSW-NB15<br>CIC-IDS2017   | 94.47 %<br>99.38 %          | No           | No                   | No                       | No                       |
| Awajan (2023)                  | 2023        | DL-IDS                                   | Custom dataset             | 93.21 %                     | No           | No                   | No                       | No                       |
| Yang et al. (2023)             | 2023        | GELU-LSTM                                | BoT-IoT                    | 99.48 %                     | No           | No                   | No                       | Yes                      |
| Bierbrauer et al. (2023)       | 2023        | TL                                       | UNSW-NB15<br>CICIDS2017    | 96.89 %<br>92.49 %          | No           | No                   | No                       | No                       |
| Rashid et al. (2023)           | 2023        | FL                                       | Edge_IIoTset               | 92.49 %                     | No           | No                   | No                       | No                       |
| Almusawi et al. (2024)         | 2024        | CNN + CBAM                               | LIVDET-2013<br>LIVDET-2015 | 98.12 %<br>97.05 %          | No           | No                   | No                       | No                       |
| GNN-IDS (Sun et al., 2024)     | 2024        | GNN                                      | CIC-IDS2017                | 94.74 %                     | No           | No                   | No                       | No                       |
| Dai et al. (2024)              | 2024        | CNN-BiLSTM-Attention                     | NSL-KDD                    | 98.3 %                      | No           | No                   | No                       | No                       |
| <b>Proposed</b>                | <b>2024</b> | <b>CNN + Channel + Spatial attention</b> | <b>Edge_IIoTset</b>        | <b>99.70 %<br/>99.85 %</b>  | <b>Yes</b>   | <b>Yes</b>           | <b>Yes</b>               | <b>Yes</b>               |

## 6. Future work

Future work could explore additional model architectures and further optimizations to continue improving detection capabilities. In particular, enhancing the explainability of model decisions is vital to building trust and transparency in industrial settings. Additionally, assessing the robustness of the proposed models against adversarial attacks will be crucial to ensure reliability in adversarial environments. Moreover, implementing and evaluating the proposed system in real-time settings will help validate its practical applicability and responsiveness in operational IoT and IIoT infrastructures. These directions will contribute to a more secure and resilient ecosystem for next-generation connected systems.

While deep learning models have demonstrated strong predictive performance, their black-box nature poses challenges in understanding how decisions are made, which can be critical for trust, debugging, and compliance in cybersecurity domains.

To address this, we have now included a discussion in the Future Work section of the manuscript highlighting our intention to incorporate model interpretability techniques. In particular, we plan to explore the use of SHAP (SHapley Additive exPlanations) to quantitatively assess the contribution of individual features toward the model's predictions. This will allow us to identify which input features are most influential in detecting specific types of attacks and could enhance model transparency and user trust.

## 7. Conclusion

The rapid expansion of IoT and IIoT technologies across industries has introduced substantial cybersecurity risks, as these interconnected devices are increasingly targeted by sophisticated cyber-attacks. Addressing these vulnerabilities is crucial, particularly in industrial settings where compromised systems can lead to significant operational disruptions and financial losses. In this study, we presented an approach to enhance attack detection accuracy in IoT and IIoT environments using

deep learning models, ResNet50 and MobileNet, augmented by attention mechanisms.

Leveraging the Edge\_IIoTset dataset, which includes both normal data and instances of 14 distinct cyber-attack types, we transformed the raw data into image representations via quantile transformation. This transformation enabled the models to capture spatial characteristics within the data effectively. By applying channel and spatial attention mechanisms, we directed the models' focus toward critical features and regions of interest, making them more sensitive to attack patterns embedded in the data.

Our findings demonstrate that incorporating attention mechanisms significantly improves the detection accuracy of ResNet50 and MobileNet across various attack types, highlighting the potential of attention-enhanced deep learning for IoT and IIoT cybersecurity. This approach not only advances the effectiveness of attack detection in resource-constrained environments but also provides a scalable, efficient solution adaptable to real-world deployment scenarios.

## CRedit authorship contribution statement

**Basharat Ahmad:** Conceptualization, Methodology, Investigation, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Yamin Li:** Writing – review & editing; **Zhaoliang Wu:** Conceptualization, Writing – review & editing; **Sadaqat Ur Rehman:** Conceptualization, Writing – review & editing; **Yongfeng Huang:** Resources, Supervision.

## Data availability

Data will be made available on request.

## Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Yongfeng Huang reports financial support was provided by the National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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